Supplementary Material for "Uncertainty-aware Dependency-based Anomaly Detection with TabPFN"

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This document provides supplementary material accompanying the paper accepted at the 2026 IEEE International Conference on Data Mining (ICDM 2026). It is organized as follows. Appendix A provides dataset statistics for all 57 ADBench datasets. Appendix B describes all 28 baseline methods. Appendix C gives implementation details and hyperparameter settings. Appendix D presents pseudocode for the training and inference procedures. Appendix E reports complete per-dataset ROC-AUC and PR-AUC results, pairwise Wilcoxon signed-rank test results, and the Friedman/Nemenyi omnibus rank analysis. Appendix F presents a detailed case study on the Wilt dataset. Appendix G reports latent-dimension sensitivity results. Appendix H reports RCS-budget sensitivity results.

Appendix A
Dataset Details

Table I summarizes all 57 ADBench datasets used in the experiments. They span a wide range of dimensionalities (3 to 1555 features), sample sizes (80 to 619326 instances), anomaly ratios, and application domains. Following the main paper, we group datasets into three dimensionality bands: low-dimensional ($d \leq 20$, 24 datasets), medium-dimensional ($20 < d \leq 100$, 18 datasets), and high-dimensional ($d > 100$, 15 datasets).

Appendix B
Baseline Methods

We compare uLEAD-TabPFN against a total of 28 baseline methods drawn from four sources. First, we include all baseline methods provided in the ADBench benchmark [1]. Second, we incorporate additional baselines evaluated by Livernoche et al. [2], which extend ADBench with recent deep learning-based, generative, and diffusion-based anomaly detection methods. Third, we additionally include two recently proposed methods that are not part of the above benchmarks, namely a PFN-based anomaly detection method, FoMo-0D [3], and a diffusion-based tabular anomaly detection approach, DDAE [4]. Fourth, we include two recent representation-learning detectors, CAIT [5] and DisentAD [6], evaluated using their official implementations.

The evaluated baselines span a broad range of anomaly detection paradigms, including conventional, deep learning-based, generative, diffusion-based, and foundation model–based approaches. Specifically, the conventional baselines include CBLOF [7], COPOD [8], ECOD [9], FeatureBagging [10], HBOS [11], Isolation Forest [12], kNN [13], LODA [14], LOF [15], MCD [16], OCSVM [17], and PCA [18]. Deep learning–based baselines include DeepSVDD [19] and DAGMM [20]. Following Livernoche et al. [2], we further include recent deep learning-based methods such as DROCC [21], GOAD [22], ICL [23], SLAD [24], and DIF [25], as well as generative and diffusion-based approaches including normalizing flows with planar flows [26], variational autoencoders (VAE) [27], generative adversarial networks (GAN) [28], and diffusion-based models using DDPMs [2], [4]. In addition, we include the PFN-based anomaly detection method FoMo-0D [3] and the two recent representation-learning detectors CAIT [5] and DisentAD [6]. Overall, our evaluation comprises 28 baseline methods.

For a fair and consistent comparison, the experimental results for all semi-supervised methods included in ADBench [1] and the additional 10 baselines introduced by Livernoche et al. [2] are taken directly from [2]. The results for FoMo-0D [3] and the diffusion-based method of [4] are reported from their respective original papers.

CAIT and DisentAD were evaluated using the author-provided codebases, under the same semi-supervised protocol as uLEAD-TabPFN: normal instances are split $50/50$ into train and test, all anomalies are placed in the test set, and ROC-AUC and PR-AUC are averaged over five seeds. Minor engineering adjustments for numerical stability and batching were applied without changing the evaluation protocol.

Appendix C
Implementation Details

uLEAD-TabPFN is designed to minimize hyperparameter tuning. The TabPFN used by uLEAD-TabPFN is kept frozen, and only lightweight components, including a linear encoder and variance networks, are trained. The primary design choice is the latent dimensionality p: for datasets with $d \leq 100$, p is set to d, while for $d > 100$, it is fixed to 100. All other parameters are fixed across datasets and seeds.

The encoder is trained for 100 epochs using Adam with no weight decay and a batch size of 1024, with the learning rate scheduled from 5×10^{-4} to 10^{-6}. For residual-scale normalization, one variance network is trained per latent dimension

Dataset	#Samples	#Features	#Anomalies	%Anomalies	Category
ALOI	49534	27	1508	3.04	Image
annthyroid	7200	6	534	7.42	Healthcare
backdoor	95329	196	2329	2.44	Network
breastw	683	9	239	34.99	Healthcare
campaign	41188	62	4640	11.27	Finance
cardio	1831	21	176	9.61	Healthcare
Cardiotocography	2114	21	466	22.04	Healthcare
celeba	202599	39	4547	2.24	Image
census	299285	500	18568	6.20	Sociology
cover	286048	10	2747	0.96	Botany
donors	619326	10	36710	5.93	Sociology
fault	1941	27	673	34.67	Physical
fraud	284807	29	492	0.17	Finance
glass	214	7	9	4.21	Forensic
Hepatitis	80	19	13	16.25	Healthcare
http	567498	3	2211	0.39	Web
InternetAds	1966	1555	368	18.72	Image
Ionosphere	351	32	126	35.90	Oryctognosy
landsat	6435	36	1333	20.71	Astronautics
letter	1600	32	100	6.25	Image
Lymphography	148	18	6	4.05	Healthcare
magic.gamma	19020	10	6688	35.16	Physical
mammography	11183	6	260	2.32	Healthcare
mnist	7603	100	700	9.21	Image
musk	3062	166	97	3.17	Chemistry
optdigits	5216	64	150	2.88	Image
PageBlocks	5393	10	510	9.46	Document
pendigits	6870	16	156	2.27	Image
Pima	768	8	268	34.90	Healthcare
satellite	6435	36	2036	31.64	Astronautics
satimage-2	5803	36	71	1.22	Astronautics
shuttle	49097	9	3511	7.15	Astronautics
skin	245057	3	50859	20.75	Image
smtp	95156	3	30	0.03	Web
SpamBase	4207	57	1679	39.91	Document
speech	3686	400	61	1.65	Linguistics
Stamps	340	9	31	9.12	Document
thyroid	3772	6	93	2.47	Healthcare
vertebral	240	6	30	12.50	Biology
vowels	1456	12	50	3.43	Linguistics
Waveform	3443	21	100	2.90	Physics
WBC	223	9	10	4.48	Healthcare
WDBC	367	30	10	2.72	Healthcare
Wilt	4819	5	257	5.33	Botany
wine	129	13	10	7.75	Chemistry
WPBC	198	33	47	23.74	Healthcare
yeast	1484	8	507	34.16	Biology
CIFAR10	5263	512	263	5.00	Image
FashionMNIST	6315	512	315	5.00	Image
MNIST-C	10000	512	500	5.00	Image
MVTec-AD	5354	512	1258	23.50	Image
SVHN	5208	512	260	5.00	Image
Agnews	10000	768	500	5.00	NLP
Amazon	10000	768	500	5.00	NLP
Imdb	10000	768	500	5.00	NLP
Yelp	10000	768	500	5.00	NLP
20news	11905	768	591	4.96	NLP

TABLE I

SUMMARY OF THE 57 ADBENCH DATASETS USED FOR EVALUATION. DATASETS ARE SORTED BY CATEGORY. LOW-DIMENSIONAL: 24 DATASETS WITH $d \leq 20$; MEDIUM-DIMENSIONAL: 18 DATASETS WITH $20 < d \leq 100$; HIGH-DIMENSIONAL: 15 DATASETS WITH $d > 100$.

for 50 epochs using Adam with a learning rate of 10^{-3} and weight decay of 10^{-4}. The RCS budget is set to 500. The weights of the dependency loss λ_{dep} and the reconstruction loss λ_{rec} are set to 1 and 0.3, respectively. The training and inference procedures are presented in Algorithms 1 and 2 in Appendix D.

Appendix D
Pseudocode

Algorithms 1 and 2 present the training and inference procedures for uLEAD-TabPFN, respectively.

Remark. $\hat{\mu}_{ij}$ denotes the conditional mean predicted for sample i and latent dimension j, i.e., $\hat{\mu}_{ij} \approx \mathbb{E}[Z_j \mid Z_{-j} = \mathbf{z}_{i,-j}, \mathcal{C}]$. $\oslash$ denotes element-wise division.

Appendix E
Additional Results

Tables II and III report detailed ROC-AUC and PR-AUC results for uLEAD-TabPFN and the baselines. Each row corresponds to one dataset and each column corresponds to one method. We report the mean ROC-AUC or PR-AUC and standard deviation over five runs (five different seeds). The value in parentheses denotes the rank of the method on that dataset (lower is better). Per-dataset ranks are computed over all compared methods; for readability, the table displays uLEAD-TabPFN and a fixed subset of high-performing baseline columns selected by aggregate mean performance over the 57 datasets.

For uLEAD-TabPFN, we additionally report two relative changes: (i) the percentage change compared to the best-performing baseline (denoted as %B), and (ii) the percentage change compared to the average baseline (denoted as %A). A positive value indicates an improvement and a negative value indicates a degradation. Both quantities, and the per-dataset ranks, are computed over the full baseline pool: all 27 ROC-AUC baselines and all 28 PR-AUC baselines (DDAE reports PR-AUC only).

A. Pairwise Wilcoxon Signed-Rank Tests

Table IV reports one-sided pairwise Wilcoxon signed-rank test p-values for the alternative that uLEAD-TabPFN outperforms each baseline, under ROC-AUC and PR-AUC. Tests are conducted separately for low-, medium-, and high-dimensional dataset groups, as well as across all 57 datasets. The "Wins" column counts the number of baselines significantly outperformed by uLEAD-TabPFN in that group, out of all 27 ROC-AUC baselines or all 28 PR-AUC baselines; the table displays a selected subset of representative baselines for readability. uLEAD-TabPFN significantly outperforms DisentAD in the medium-, high-, and all-dataset settings, and its significant advantages over the remaining methods are most frequent in the high-dimensional setting. Against CAIT, the test does not reach significance in any group: CAIT is stronger on low- and medium-dimensional data, while in the high-dimensional setting uLEAD-TabPFN is ahead on average but the difference is not statistically significant (p=0.08 for ROC-AUC and p=0.06 for PR-AUC).

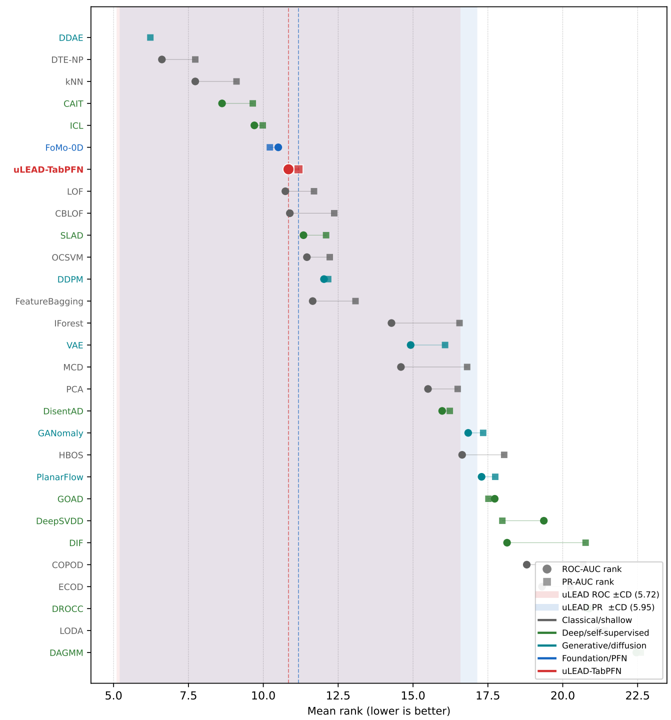

Fig. 1. Critical-difference rank plot for the Friedman/Nemenyi analysis over all 57 datasets. Lower mean rank is better. Horizontal bars connect groups of methods that are *not* significantly different from each other under the Nemenyi post-hoc test at α=0.05. Methods not connected to uLEAD-TabPFN by a bar are significantly different from it. uLEAD-TabPFN is highlighted; no method ranks significantly above it.

B. Friedman/Nemenyi Omnibus Rank Analysis

The Wilcoxon results above provide dimensionality-stratified pairwise evidence. As a complementary omnibus test, the Friedman test evaluates whether all methods have equal rank distributions over the 57 datasets; a significant result licenses post-hoc pairwise comparisons. The Nemenyi post-hoc test then compares method ranks while explicitly accounting for multiple comparisons across all method pairs.

Table V summarises the results. The Friedman test strongly rejects the null hypothesis of equal rank distributions for both ROC-AUC ($p \approx 3.76 \times 10^{-75}$, 28 methods) and PR-AUC ($p \approx 1.91 \times 10^{-78}$, 29 methods). Under the subsequent Nemenyi post-hoc test at α=0.05, uLEAD-TabPFN achieves mean ranks of 10.84 (ROC-AUC) and 11.18 (PR-AUC) out of 28 and 29 methods, respectively. No compared method ranks significantly *above* uLEAD-TabPFN, while 11 methods rank significantly *below* it for both metrics. This result supports the conclusion that uLEAD-TabPFN belongs to the statistically leading group overall. The dimensionality-stratified Wilcoxon results in Table IV provide complementary pairwise evidence and show that the strongest significant gains occur in the high-dimensional regime ($d > 100$).

Figure 1 shows the critical-difference rank plot for the all-dataset Friedman/Nemenyi analysis.

TABLE II

ROC-AUC comparison on ADBench datasets. We report mean±std over five runs (five different seeds). Numbers in parentheses denote the per-dataset rank among all 28 methods (uLEAD-TabPFN and all 27 ROC-AUC baselines). uLEAD-TabPFN additionally reports the relative change compared to the best baseline (%B) and the average baseline (%A). Per-dataset ranks and relative changes are computed over the full baseline pool; for readability, the table displays uLEAD-TabPFN and a fixed subset of high-performing baseline columns selected by aggregate mean ROC-AUC over the 57 datasets.

Dataset	uLEAD-TabPFN	CAIT	DTE-NP	kNN	FoMo-0D	ICL	LOF	CBLOF	DDPM	SLAD	FeatureBagging
skin	95.63±2.31(4) [-4.03%B, +26.23%A]	99.64±0.09(1)	98.86±0.46(3)	99.49±0.08(2)	70.60±0.88(19)	6.58±0.63(28)	86.34±1.77(14)	91.82±0.21(5)	88.74±4.40(11)	91.05±2.61(7)	78.39±0.91(16)
smtp	91.33±1.65(9) [-4.30%B, +7.99%A]	84.65±1.40(18)	92.98±2.91(5)	92.43±2.70(6)	83.04±4.69(20)	74.36±7.07(25)	93.42±2.48(4)	87.28±5.65(13)	95.43±1.26(1)	92.15±1.87(7)	84.81±3.66(16)
http	98.25±1.40(19) [-1.75%B, +6.93%A]	99.83±0.07(12)	99.98±0.03(4.5)	100.00±0.00(2)	99.92±0.05(10)	98.24±3.45(20)	99.98±0.03(4.5)	99.93±0.01(9)	100.00±0.00(2)	99.91±0.08(11)	92.10±0.57(23)
wilt	97.47±0.94(1) [+0.35%B, +79.26%A]	87.02±2.52(3)	62.91±5.59(11)	63.66±0.00(10)	97.13±0.00(2)	76.42±3.46(5)	68.81±0.00(9)	42.90±1.14(18)	71.66±0.81(8)	61.76±0.11(12)	73.35±10.50(7)
vertebral	52.78±6.69(10) [-39.47%B, +3.42%A]	50.87±5.26(11)	54.30±15.47(9)	57.67±3.58(6)	87.20±2.58(1)	79.19±5.06(2)	64.30±1.31(4)	54.43±1.97(8)	70.67±6.28(3)	44.96±0.25(19)	64.13±2.15(5)
annthyroid	81.60±5.96(18) [-12.58%B, -2.22%A]	91.04±0.74(5)	92.90±0.25(3)	92.81±0.00(4)	85.17±0.00(17)	81.11±1.07(19)	88.63±0.00(12)	90.14±1.08(8)	88.82±1.34(11)	93.34±0.36(1)	88.95±1.61(9)
mammography	71.73±2.77(25) [-20.89%B, -12.83%A]	89.66±0.55(4)	87.62±0.09(9)	87.58±0.00(10)	69.11±1.21(28)	71.87±9.11(24)	85.52±0.00(13)	84.74±0.02(15)	81.01±2.04(18)	74.51±0.67(21)	86.31±0.35(11)
thyroid	88.18±3.89(28) [-10.89%B, -8.32%A]	97.56±0.55(12)	98.63±0.04(4)	98.68±0.00(2)	96.84±0.00(14)	95.40±0.98(17)	92.72±0.00(25)	98.54±0.06(8)	97.95±0.22(11)	95.27±0.24(18)	93.15±1.52(24)
glass	87.59±2.42(9) [-11.92%B, +9.37%A]	86.13±2.15(10)	89.64±3.54(5)	92.04±1.12(4)	98.30±0.53(2)	99.44±0.58(1)	88.82±1.98(7)	89.35±1.48(6)	66.67±13.63(25)	86.04±5.69(11)	88.49±1.71(8)
yeast	57.41±1.07(1) [+9.29%B, +24.96%A]	43.79±2.10(21)	44.58±0.33(20)	44.74±0.00(18)	49.91±0.00(5)	48.98±2.39(8)	45.79±0.00(15)	50.40±0.08(4)	49.13±2.85(6)	48.69±0.11(9)	46.40±1.33(14)
pima	64.75±3.46(19) [-20.55%B, -5.11%A]	72.46±1.47(10)	81.50±2.57(1)	76.94±1.87(4)	80.46±1.48(2)	79.68±3.06(3)	70.53±2.19(15)	72.94±1.07(9)	70.27±2.28(16)	60.62±4.21(22)	71.93±2.22(13)
stamps	84.79±3.37(22) [-13.36%B, -3.52%A]	94.16±1.07(6)	97.87±0.37(1)	95.89±1.44(4)	97.52±0.39(2)	96.68±1.09(3)	93.74±2.36(7)	93.41±1.66(10)	91.84±4.15(15)	81.97±8.29(23)	94.21±2.26(5)
wbc	79.98±5.19(26) [-19.84%B, -14.61%A]	95.81±1.85(22)	99.54±0.28(5)	99.12±0.19(13)	99.61±0.31(4)	99.66±0.36(2)	80.52±4.47(25)	98.31±1.31(16)	99.20±0.39(11)	99.78±0.18(1)	58.05±18.08(27)
breastw	87.14±1.20(25) [-12.45%B, -6.32%A]	98.71±0.35(14)	99.28±0.12(5)	99.05±0.26(12)	99.15±0.23(9)	98.28±0.45(17)	88.91±6.80(24)	99.11±0.23(11)	98.70±0.43(15)	99.53±0.13(1)	59.07±15.44(26)
shuttle	99.23±0.12(18) [-0.75%B, +5.49%A]	99.79±0.08(9)	99.93±0.02(2)	99.91±0.00(4.5)	99.82±0.09(7)	99.92±0.04(3)	99.98±0.00(1)	99.72±0.02(11)	99.91±0.01(4.5)	99.90±0.01(6)	86.89±8.22(23)
magic.gamma	83.46±1.09(6) [-5.80%B, +13.01%A]	88.60±0.49(1)	83.57±0.76(5)	83.27±0.00(8)	84.77±0.14(3)	75.56±0.42(12)	83.40±0.00(7)	75.81±0.00(11)	85.97±1.08(2)	72.00±0.01(17)	84.19±0.72(4)
pageblocks	93.45±1.51(2) [-3.09%B, +8.42%A]	96.43±0.38(1)	89.32±0.26(9)	89.65±0.00(8)	86.58±0.00(18)	88.39±0.77(11)	91.30±0.00(5)	91.23±0.11(6)	86.93±0.42(17)	87.86±0.01(14)	91.11±0.30(7)
donors	99.99±0.01(1) [+0.08%B, +17.12%A]	99.17±0.41(6)	99.26±0.29(5)	99.49±0.06(4)	99.91±0.02(2)	99.90±0.05(3)	96.97±0.24(7)	93.47±0.23(10)	82.50±1.83(19)	88.51±5.51(17)	95.21±1.72(9)
cover	82.04±5.04(18) [-17.28%B, -0.27%A]	98.92±0.37(4)	97.73±0.55(6)	97.54±0.15(7)	99.12±0.29(3)	89.34±4.02(15)	99.18±0.10(1)	94.04±0.28(13)	98.35±0.66(5)	73.97±13.80(21)	99.16±0.63(2)
vowels	97.97±1.01(2) [-1.00%B, +43.95%A]	98.96±0.46(1)	81.42±1.44(10)	82.21±0.00(9)	81.29±0.00(11)	85.10±2.13(7)	86.30±0.00(5)	78.72±4.01(12)	86.38±1.94(4)	85.02±0.02(8)	85.32±1.16(6)
wine	95.73±5.14(14) [-4.27%B, +4.71%A]	99.37±0.56(6)	99.44±0.97(5)	99.19±0.22(7)	99.94±0.12(2)	99.87±0.29(3)	98.36±0.38(8)	97.75±0.59(11)	99.61±0.09(4)	100.00±0.00(1)	97.93±0.87(9)
pendigits	97.91±1.20(8) [-1.96%B, +9.84%A]	99.60±0.15(3)	99.61±0.17(2)	99.87±0.00(1)	98.11±0.00(6.5)	96.71±0.83(10)	99.05±0.00(5)	96.67±0.07(11)	98.11±0.23(6.5)	94.61±0.41(13)	99.50±0.11(4)
lymphography	88.08±9.11(25) [-11.92%B, -7.46%A]	86.85±7.05(26)	99.93±0.09(6)	99.93±0.08(6)	99.93±0.11(6)	100.00±0.00(2)	98.21±0.75(22)	99.83±0.02(12)	99.94±0.09(4)	100.00±0.01(2)	96.61±2.84(23)
hepatitis	75.73±3.07(20) [-24.22%B, -9.64%A]	62.62±7.02(27)	93.22±3.90(9)	96.46±1.46(6)	99.88±0.24(3)	99.94±0.13(1)	66.92±7.01(26)	86.26±2.35(12)	97.74±1.18(5)	99.93±0.15(2)	67.76±6.50(25)
cardiotocography	71.07±3.69(11) [-10.38%B, +9.47%A]	72.60±2.88(9)	63.76±1.88(17)	62.11±0.00(20)	72.15±0.00(10)	54.20±1.80(24)	64.49±0.00(16)	67.61±2.21(13)	54.54±2.96(23)	47.32±0.31(26)	63.64±2.40(18)
waveform	62.00±5.28(20) [-19.43%B, -5.84%A]	71.33±3.23(9)	74.48±0.00(5)	75.21±0.00(4)	71.02±0.00(10)	68.68±3.66(13)	76.00±0.00(2)	72.93±0.90(6)	62.17±2.64(19)	48.92±0.08(28)	76.95±1.07(1)
cardio	84.84±1.39(20) [-12.13%B, -2.98%A]	91.48±1.25(15)	91.80±0.59(14)	92.00±0.00(13)	94.27±0.00(6)	80.01±2.12(24)	92.21±0.00(11)	93.49±1.47(8)	86.94±1.96(18)	83.05±1.10(21)	92.12±0.56(12)
fault	75.13±0.76(2) [-2.30%B, +31.41%A]	76.90±0.65(1)	58.64±0.65(14)	58.73±0.00(13)	61.82±0.00(5)	60.63±0.50(7)	47.42±0.00(28)	59.00±1.29(11)	61.09±1.16(6)	63.93±0.19(3)	48.32±0.95(27)
aloi	58.47±0.56(1) [+4.37%B, +14.23%A]	56.02±0.46(2)	51.19±0.46(12)	51.04±0.00(14)	54.05±0.30(4)	47.50±0.98(28)	48.76±0.00(24)	53.69±0.15(8)	49.91±0.35(20)	50.76±0.29(17)	49.07±0.53(23)
fraud	95.41±0.61(5) [-0.24%B, +5.54%A]	95.35±0.16(7)	95.64±1.05(1)	95.43±1.04(4)	93.91±2.81(17)	92.78±1.46(20)	94.35±1.41(15)	94.91±1.08(10)	93.65±0.92(18)	94.38±1.01(14)	94.83±1.56(12)
wdbc	97.77±1.17(20) [-2.01%B, +3.46%A]	99.02±0.43(14)	99.52±0.38(5)	99.05±0.27(13)	99.77±0.25(2)	99.78±0.27(1)	99.62±0.21(4)	98.73±0.43(17.5)	99.30±0.17(9)	99.49±0.25(6)	99.63±0.18(3)
letter	93.24±1.60(2) [-0.42%B, +126.73%A]	93.64±0.58(1)	34.38±0.98(19)	35.43±0.00(18)	40.90±0.00(10)	42.68±1.17(9)	44.83±0.00(8)	33.24±0.66(21)	38.05±1.12(13)	36.80±0.41(14)	44.84±1.00(7)
ionosphere	95.62±1.16(12) [-3.39%B, +5.83%A]	98.19±0.25(3)	97.77±1.39(4)	97.44±0.98(5)	96.59±1.25(9)	98.98±0.32(1)	94.29±2.20(16)	96.78±1.50(8)	94.60±0.85(14)	98.21±0.60(2)	94.47±2.10(15)
wpbc	55.76±1.97(17) [-42.28%B, -10.14%A]	52.33±1.95(21)	83.16±13.46(3)	63.67±2.34(8)	76.10±2.11(6)	96.61±1.17(1)	57.36±2.01(14)	59.57±1.98(12)	66.49±3.17(7)	95.53±2.21(2)	56.79±1.86(15)
satellite	82.72±0.57(6) [-5.45%B, +8.66%A]	85.93±0.97(2)	82.11±0.67(8)	82.24±0.00(7)	84.98±0.00(5)	85.15±0.48(4)	80.30±0.00(9)	73.20±0.92(19)	77.70±0.53(12)	87.49±0.10(1)	80.11±0.11(10)
satimage-2	99.66±0.30(6) [-0.26%B, +1.73%A]	99.71±0.11(4)	99.67±0.00(5)	99.71±0.00(3)	98.40±0.00(19)	99.48±0.22(9)	99.38±0.00(12)	99.42±0.01(11)	99.62±0.16(7)	99.77±0.00(2)	99.47±0.03(10)
landsat	67.32±0.52(6) [-9.60%B, +17.43%A]	74.47±1.90(1)	68.20±1.75(5)	68.25±0.00(4)	69.53±0.00(3)	65.13±0.44(9)	66.58±0.00(7)	57.21±0.26(13)	51.37±1.00(21)	65.03±0.18(10)	66.38±0.17(8)
celeba	74.43±0.94(11) [-11.78%B, +8.02%A]	73.45±3.27(12)	70.40±0.37(17)	73.14±0.72(13)	74.81±0.61(10)	72.21±0.82(14)	43.73±0.75(28)	79.28±1.30(5)	78.56±1.99(6)	67.43±1.64(20)	46.89±1.80(26)
spambase	80.48±1.07(15) [-5.52%B, +5.54%A]	82.68±1.24(6)	83.74±0.69(3)	83.36±0.00(5)	78.21±0.00(16)	83.53±0.45(4)	73.23±0.00(19)	81.52±0.55(11)	64.54±0.80(27)	84.86±0.20(2)	69.64±2.09(23)
campaign	74.71±1.11(14) [-7.67%B, +5.59%A]	77.91±2.28(6)	78.79±0.25(2)	78.48±0.00(4)	65.30±0.70(22)	80.92±0.79(1)	70.55±0.00(17)	77.05±0.31(11)	74.51±0.42(15)	76.75±0.16(13)	69.10±3.85(20)
optdigits	85.48±5.00(10) [-12.04%B, +20.14%A]	94.09±2.01(6)	94.28±1.67(5)	93.72±0.00(7)	85.28±0.00(11)	97.18±0.80(1)	96.65±0.00(2)	83.52±1.69(13)	90.76±2.12(8)	95.28±0.17(4)	96.27±0.49(3)
mnist	88.30±1.60(15.5) [-9.03%B, +8.20%A]	97.07±0.19(1)	94.02±0.42(2)	93.85±0.00(3)	91.27±0.00(6)	90.11±1.13(11)	92.93±0.00(4)	91.10±0.23(7)	87.27±3.22(17)	89.73±0.44(13)	92.55±0.40(5)
musk	100.00±0.00(8.5) [+0.00%B, +4.44%A]	100.00±0.00(8.5)	100.00±0.00(8.5)	100.00±0.00(8.5)	99.62±0.43(21)	99.37±0.63(22)	100.00±0.00(8.5)	100.00±0.00(8.5)	100.00±0.00(8.5)	100.00±0.00(8.5)	100.00±0.00(8.5)
backdoor	96.07±0.46(1) [+0.78%B, +28.30%A]	93.51±0.31(7)	93.31±1.70(8)	93.75±0.53(5)	78.59±17.11(14)	93.62±0.69(6)	95.33±0.25(2)	69.65±5.23(18)	80.93±0.56(13)	50.00±0.00(26)	94.83±0.62(3)
speech	60.53±2.26(1) [+15.76%B, +46.91%A]	49.95±2.66(4)	41.37±0.00(10.5)	36.36±0.00(26)	40.10±1.75(12)	48.86±2.72(7)	37.53±0.00(17)	35.88±0.15(28)	36.96±0.86(20)	41.37±0.94(10.5)	37.48±0.37(18)
census	70.49±1.23(8) [-4.92%B, +14.85%A]	61.37±1.08(15)	72.10±0.40(3)	72.26±0.29(2)	60.54±4.96(16)	70.56±0.35(5)	58.46±1.06(19)	70.84±0.28(4)	70.15±0.23(9)	57.91±10.84(20)	55.92±1.00(21)
CIFAR10	62.16±7.07(20) [-14.13%B, -2.63%A]	72.39±0.43(1)	67.82±0.00(7)	67.53±0.00(9.5)	64.53±0.62(16)	63.60±0.83(18)	70.30±0.00(4)	67.87±0.21(6)	67.91±0.13(5)	66.60±0.10(14)	70.31±0.21(3)
SVHN	59.79±2.64(17) [-6.48%B, +2.10%A]	63.63±1.12(3)	62.13±0.00(4)	61.69±0.00(6)	59.93±0.64(16)	61.70±0.50(5)	63.82±0.00(2)	60.97±0.17(10)	61.37±0.08(7)	60.87±0.07(11)	63.93±0.14(1)
MVTec-AD	78.31±10.98(14) [-18.95%B, -0.09%A]	96.62±0.47(1)	89.66±1.51(5)	81.51±1.82(9)	75.31±1.26(22)	94.75±0.84(3)	80.36±2.14(12)	79.96±2.02(13)	78.00±1.99(15)	93.24±1.09(4)	80.45±2.20(11)
FashionMNIST	88.62±6.53(9) [-3.33%B, +9.12%A]	87.92±0.38(13)	90.14±0.00(4)	89.87±0.00(6)	86.66±0.66(17)	90.56±0.27(3)	91.60±0.00(2)	89.10±0.15(8)	88.52±0.07(10)	89.94±0.04(5)	91.67±0.11(1)
MNIST-C	86.71±11.46(3) [-0.71%B, +20.21%A]	49.21±0.67(27)	84.74±0.00(5)	84.11±0.00(6)	77.28±1.69(15)	85.62±0.52(4)	87.21±0.00(2)	81.11±0.09(8)	80.12±0.14(9)	83.51±0.11(7)	87.33±0.15(1)
yelp	68.18±0.43(2) [-0.70%B, +15.67%A]	55.58±2.01(22)	68.66±0.00(1)	68.07±0.00(3)	60.29±5.14(11)	55.79±0.50(21)	67.20±0.00(4)	63.80±0.07(7)	59.30±0.09(15)	54.88±0.19(23)	67.06±0.12(5)
imdb	52.25±0.73(2) [-0.17%B, +5.11%A]	50.93±1.00(9)	50.43±0.00(11)	50.08±0.00(12)	50.84±6.24(10)	52.34±0.49(1)	49.57±0.00(16)	49.94±0.01(14.5)	47.91±0.10(26)	51.26±0.10(6)	49.53±0.10(17.5)
agnews	72.44±5.52(3) [-2.95%B, +22.77%A]	59.13±3.10(12)	67.98±0.00(4.5)	67.05±0.00(6)	60.96±3.69(9)	62.55±0.47(8)	74.58±0.00(2)	62.82±0.07(7)	57.82±0.12(16)	58.43±0.07(14.5)	74.64±0.09(1)
20news	61.37±9.36(4) [-3.66%B, +8.28%A]	63.70±3.58(1)	59.97±0.74(8)	57.36±0.66(11)	56.93±1.46(13)	61.30±1.05(5)	60.19±0.62(7)	57.10±1.23(12)	54.88±0.52(18)	59.47±0.80(10)	60.25±0.70(6)
amazon	62.01±0.41(1) [+1.96%B, +16.45%A]	55.43±0.66(13)	60.82±0.00(2)	60.58±0.00(3)	54.74±4.18(17)	54.21±0.22(18)	57.88±0.00(7)	58.17±0.12(5)	55.09±0.10(15)	52.01±0.06(22)	57.94±0.04(6)
internetads	90.76±0.32(1) [+8.18%B, +41.36%A]	83.90±3.56(2)	69.96±2.22(10)	68.08±0.00(12)	55.64±7.12(22)	72.20±0.55(5)	71.72±0.00(6)	65.16±0.08(18)	65.76±0.06(15)	75.94±0.11(3)	71.38±2.32(7)

TABLE III

PR-AUC COMPARISON ON ADBENCH DATASETS. WE REPORT MEAN±STD OVER FIVE RUNS (FIVE DIFFERENT SEEDS). NUMBERS IN PARENTHESES DENOTE THE PER-DATASET RANK AMONG ALL 29 METHODS (uLEAD-TABPFN AND ALL 28 PR-AUC BASELINES, AS DDAE REPORTS PR-AUC ONLY). uLEAD-TABPFN ADDITIONALLY REPORTS THE RELATIVE CHANGE COMPARED TO THE BEST BASELINE (%B) AND THE AVERAGE BASELINE (%A). PER-DATASET RANKS AND RELATIVE CHANGES ARE COMPUTED OVER THE FULL BASELINE POOL; FOR READABILITY, THE TABLE DISPLAYS uLEAD-TABPFN AND A FIXED SUBSET OF HIGH-PERFORMING BASELINE COLUMNS SELECTED BY AGGREGATE MEAN PR-AUC OVER THE 57 DATASETS.

Dataset	uLEAD-TabPFN	DDAE	CAIT	DTE-NP	FoMo-0D	kNN	ICL	DDPM	LOF	SLAD	OCSVM
skin	86.77±8.03(5) [-12.40%B, +44.57%A]	92.23±0.01(4)	99.06±0.31(1)	94.78±2.33(3)	51.04±1.72(19)	98.24±0.41(2)	32.46±1.00(26)	76.37±5.79(7)	61.68±1.85(16)	78.73±7.88(6)	66.31±0.51(11)
smtp	33.48±6.18(16) [-50.77%B, +8.01%A]	46.23±0.08(13)	50.43±14.44(6)	50.20±6.39(7)	38.18±9.90(15)	50.53±5.92(5)	3.81±3.83(22)	40.81±13.36(14)	48.09±7.38(12)	50.60±6.30(4)	64.51±11.88(2)
http	48.70±1.46(20) [-51.30%B, -23.04%A]	99.25±0.01(4)	82.91±6.84(13)	97.10±4.04(6)	90.58±5.01(9)	100.00±0.00(1)	70.82±42.06(14)	99.96±0.06(2)	97.12±3.92(5)	88.09±7.46(12)	99.88±0.26(3)
wilt	84.63±5.63(1) [+9.26%B, +476.64%A]	14.79±0.02(10)	30.96±3.61(3)	12.20±1.60(12)	77.46±0.00(2)	12.25±0.00(11)	28.94±3.31(4)	17.24±0.46(7)	15.74±0.00(9)	12.17±0.04(13)	7.12±0.00(26)
vertebral	22.33±2.84(16) [-67.82%B, -14.86%A]	27.75±0.04(7)	21.59±2.07(18)	25.21±8.90(10)	69.39±3.97(1)	26.11±2.49(8)	58.75±7.30(2)	35.84±9.27(3)	33.87±4.30(4)	19.87±4.37(23)	22.23±2.01(17)
annthyroid	45.90±3.90(22) [-34.97%B, -14.43%A]	56.88±0.02(14)	60.97±3.02(9)	68.15±0.38(2)	48.81±0.00(19)	68.07±0.00(3)	45.83±2.16(23)	62.88±2.59(7)	53.53±0.00(17)	70.58±0.92(1)	60.11±0.00(10)
mammography	28.64±3.46(17) [-48.12%B, -10.87%A]	35.21±0.07(14)	47.61±2.55(3)	42.09±0.86(5)	36.28±1.26(13)	41.27±0.00(8)	17.11±3.75(27)	19.93±3.92(24)	34.07±0.00(15)	18.98±1.05(25)	40.52±0.00(10)
thyroid	56.84±9.20(24) [-30.87%B, -17.21%A]	67.34±0.03(17)	71.17±7.44(15)	81.03±0.31(5)	67.00±0.00(18)	80.94±0.00(6)	51.51±12.75(27)	82.22±0.87(1)	60.57±0.00(23)	74.07±0.84(14)	78.92±0.00(10)
glass	29.61±8.62(14) [-67.94%B, -12.48%A]	31.47±0.07(11)	29.43±7.41(15)	37.38±15.51(8)	83.60±4.63(2)	42.32±8.47(5)	92.35±8.32(1)	31.21±11.30(12)	38.12±9.91(7)	41.15±3.98(6)	26.76±7.68(17)
yeast	55.76±1.16(2) [-17.77%B, +12.68%A]	67.81±0.02(1)	48.08±1.52(21)	48.12±0.49(20)	51.00±0.00(6)	48.26±0.00(19)	49.55±1.36(13)	51.05±1.65(5)	48.94±0.00(18)	50.56±0.08(9)	47.95±0.00(22)
pima	65.24±4.65(20) [-20.91%B, -5.21%A]	82.49±0.02(1)	71.54±2.23(10)	79.70±2.44(2)	79.27±1.74(3)	75.37±2.99(6)	78.63±1.94(4)	71.18±2.06(13.5)	68.40±3.79(18)	63.03±4.45(23)	71.98±3.41(9)
stamps	51.52±9.84(20) [-42.35%B, -11.27%A]	73.81±0.12(4)	62.59±4.91(10)	82.47±4.08(2)	89.37±3.50(1)	71.68±8.35(5)	79.54±5.44(3)	64.74±12.87(9)	64.84±8.17(8)	50.61±13.26(21)	64.91±7.95(7)
wbc	53.93±5.84(26) [-45.05%B, -31.49%A]	87.83±0.03(15)	67.13±12.98(23)	96.10±2.17(4)	96.46±2.49(3)	92.01±3.96(12)	95.12±5.14(5)	93.84±2.45(8)	24.89±3.49(27)	98.14±1.35(1)	97.15±0.77(2)
breastw	79.90±3.24(26) [-19.71%B, -14.01%A]	99.51±0.00(1)	98.31±0.49(16)	99.19±0.14(6.5)	99.03±0.34(12)	98.92±0.32(13)	96.79±1.61(20)	98.60±0.51(15)	80.01±10.09(25)	99.47±0.21(3)	99.35±0.21(5)
shuttle	98.53±0.31(7) [-1.22%B, +13.42%A]	99.11±0.00(4)	98.99±0.19(5)	98.14±0.48(8)	99.36±0.17(3)	97.86±0.00(13)	99.72±0.14(2)	97.91±0.26(12)	99.75±0.00(1)	98.04±0.01(10)	97.67±0.00(14)
magic.gamma	85.62±1.41(9) [-8.50%B, +8.97%A]	93.57±0.00(1)	90.80±0.34(2)	86.15±0.74(7)	87.54±0.18(4)	85.86±0.00(8)	81.33±0.57(11)	87.97±0.84(3)	86.36±0.00(6)	77.34±0.01(16)	79.16±0.00(14)
pageblocks	81.89±3.04(3) [-3.85%B, +32.99%A]	85.04±0.01(2)	85.17±2.42(1)	67.45±0.10(11)	62.67±0.00(16)	67.60±0.00(10)	68.11±2.30(9)	62.10±0.95(17)	71.07±0.00(6)	64.70±0.79(12)	64.25±0.00(13)
donors	99.88±0.11(1) [+1.16%B, +99.35%A]	98.73±0.01(2)	86.96±5.82(6)	85.55±4.56(7)	98.43±0.48(3)	89.09±0.94(5)	98.35±0.87(4)	26.66±2.91(25)	63.39±1.89(10)	46.18±9.80(13)	42.71±0.86(15)
cover	36.55±4.25(9) [-57.03%B, +20.41%A]	79.50±0.02(3)	85.07±2.89(1)	59.97±10.57(7)	72.57±8.38(6)	55.79±3.74(8)	34.48±16.39(10)	73.27±3.36(5)	82.92±2.19(2)	6.96±5.18(22)	22.28±1.01(14)
vowels	79.51±8.49(3) [-15.99%B, +207.01%A]	94.64±0.02(1)	87.57±5.75(2)	31.59±1.59(9)	24.10±0.00(13)	30.21±0.00(10)	27.39±5.75(12)	42.72±4.80(5)	33.09±0.00(7)	39.23±1.65(6)	27.43±0.00(11)
wine	82.77±18.44(13) [-17.23%B, +10.06%A]	79.15±0.06(15)	96.80±2.70(5)	96.80±5.58(6)	99.62±0.77(2)	95.11±1.81(7)	98.26±3.89(3)	97.65±0.72(4)	89.95±2.64(8)	100.00±0.00(1)	88.68±2.29(9.5)
pendigits	71.43±11.94(7) [-26.35%B, +53.02%A]	92.65±0.01(2)	88.36±4.60(4)	91.89±3.30(3)	66.33±0.00(9)	96.99±0.00(1)	66.41±7.58(8)	61.14±4.73(10)	78.55±0.00(6)	35.35±0.15(19)	51.78±0.00(12)
lymphography	66.31±19.50(27) [-33.69%B, -25.17%A]	92.52±0.08(20)	72.59±3.87(26)	99.34±0.93(3.5)	99.15±1.26(7)	99.17±0.94(6)	100.00±0.00(1.5)	99.30±1.02(5)	84.16±4.27(23)	99.34±0.12(3.5)	100.00±0.00(1.5)
hepatitis	58.35±5.37(17) [-41.55%B, -14.35%A]	54.92±0.07(22)	38.06±3.91(28)	82.32±10.26(9)	99.45±1.10(3)	90.31±4.36(6)	99.83±0.38(1)	95.14±1.34(5)	43.67±10.79(27)	99.79±0.47(2)	77.63±3.49(10)
cardiotocography	63.69±2.92(9) [-8.61%B, +9.51%A]	68.19±0.02(5)	64.77±3.21(8)	58.68±1.14(16)	63.55±0.00(10)	57.43±0.00(17)	48.66±3.84(26)	51.31±1.98(23)	57.32±0.00(18)	49.37±0.19(25)	66.19±0.00(7)
waveform	20.36±4.07(8) [-33.59%B, +45.10%A]	21.00±0.04(7)	14.89±2.87(11)	27.87±0.00(3)	9.95±0.00(16)	27.00±0.00(4)	18.63±4.08(10)	9.31±1.05(18)	30.66±0.00(1)	5.31±0.01(29)	10.91±0.00(14)
cardio	64.76±5.37(23) [-24.92%B, -7.57%A]	79.60±0.02(7)	73.93±2.62(14)	77.41±0.85(11)	78.92±0.00(8)	77.22±0.00(12)	47.91±11.47(27)	69.28±0.91(19)	70.15±0.00(17)	69.94±0.98(18)	83.59±0.00(4)
fault	73.05±0.79(3) [-17.49%B, +20.00%A]	88.53±0.01(1)	77.08±1.23(2)	62.17±0.14(10)	63.08±0.00(8)	61.98±0.00(12)	63.18±0.70(7)	64.75±0.69(5)	50.44±0.00(29)	66.69±0.19(4)	61.12±0.00(14)
aloi	10.39±0.41(2) [-7.18%B, +60.21%A]	8.71±0.00(3)	11.19±0.30(1)	6.05±0.06(17)	6.93±0.05(6)	6.02±0.00(18)	5.50±0.13(28)	5.97±0.06(20)	6.54±0.00(9)	5.99±0.04(19)	6.52±0.00(11)
fraud	71.63±5.21(1) [+3.50%B, +78.61%A]	48.24±0.04(13)	52.55±5.36(10)	42.10±7.63(15)	55.13±13.77(7)	38.68±7.19(16)	53.88±8.78(9)	69.21±3.46(2)	55.09±8.19(8)	44.97±5.43(14)	29.64±5.04(21)
wdbc	69.98±3.65(20) [-26.80%B, -3.14%A]	65.13±0.08(22)	80.64±9.96(14)	90.47±7.91(5)	93.82±8.30(2)	82.03±3.28(13)	95.60±6.12(1)	84.30±4.44(9)	93.64±3.06(4)	89.08±6.97(6)	87.44±5.59(7)
letter	67.90±5.78(2) [-17.26%B, +357.60%A]	82.06±0.05(1)	66.62±3.31(3)	8.57±0.13(20)	9.33±0.00(13)	8.70±0.00(19)	12.80±1.17(7)	9.53±0.51(12)	11.26±0.00(9)	8.93±0.05(15.5)	8.26±0.00(23)
ionosphere	96.34±1.16(14) [-2.75%B, +5.14%A]	98.41±0.00(3)	98.65±0.20(2)	98.22±1.02(4)	97.75±0.76(7)	97.95±0.69(6)	99.06±0.32(1)	96.43±0.50(13)	94.58±1.57(18)	95.38±0.51(15.5)	97.45±0.53(9)
wpbc	38.40±2.32(23) [-57.00%B, -21.91%A]	47.16±0.03(8)	39.65±1.56(21)	69.02±13.91(5)	67.36±4.34(6)	46.11±2.74(10)	89.31±5.37(1)	54.61±3.75(7)	41.20±2.62(15)	87.45±6.21(2)	40.88±3.04(17)
satellite	86.73±0.41(6) [-6.28%B, +6.60%A]	92.54±0.00(1)	87.87±0.84(4)	85.83±0.72(10)	88.05±0.00(3)	86.01±0.00(8)	87.62±0.24(5)	85.09±0.18(12)	85.86±0.00(9)	88.64±0.07(2)	80.90±0.00(17)
satimage-2	93.41±2.80(11) [-4.98%B, +9.55%A]	93.29±0.03(12)	91.05±7.59(17)	96.16±0.00(5)	92.78±0.00(14)	96.69±0.00(4)	94.70±1.19(8)	88.05±5.10(20)	88.46±0.00(19)	95.44±0.23(7)	96.92±0.00(2)
landsat	55.01±0.34(7) [-15.68%B, +23.79%A]	65.24±0.01(1)	57.81±3.17(6)	54.52±4.05(9)	58.24±0.00(5)	54.85±0.00(8)	53.12±1.57(10)	34.83±0.91(24)	61.37±0.00(3)	45.10±0.17(13)	37.01±0.00(20)
celeba	10.50±0.51(16) [-49.88%B, -10.29%A]	11.49±0.01(13)	8.64±1.08(20)	10.65±0.49(15)	8.39±0.38(21)	11.92±0.50(11)	9.74±0.68(17)	18.03±2.60(6)	3.61±0.15(28)	2.90±0.92(29)	20.27±0.90(3)
spambase	80.69±1.87(17) [-14.08%B, +1.58%A]	93.91±0.00(1)	87.77±1.04(3)	83.65±0.58(7)	80.99±0.00(16)	83.32±0.00(9)	86.78±0.59(4)	72.89±0.42(25)	72.71±0.00(26)	85.64±0.14(5)	82.19±0.00(10)
campaign	46.22±1.73(16) [-13.66%B, +9.59%A]	53.53±0.01(1)	48.33±2.35(13)	49.95±0.67(3)	34.24±0.47(23)	49.04±0.00(7)	48.90±0.98(8)	48.87±0.29(9)	40.24±0.00(19)	48.11±0.12(14)	49.43±0.00(6)
optdigits	18.96±7.15(13) [-62.78%B, -4.71%A]	39.69±0.02(6)	50.36±12.19(2)	31.75±4.86(9)	31.88±0.00(8)	29.11±0.00(10)	50.94±8.59(1)	25.56±4.25(11)	43.63±0.00(3)	36.30±0.94(7)	6.92±0.00(19)
mnist	64.59±4.98(14) [-24.32%B, +15.02%A]	77.11±0.02(2)	85.35±2.09(1)	73.68±1.31(3)	57.97±0.00(18)	72.72±0.00(4)	68.45±1.63(7)	62.42±4.39(16)	70.97±0.00(5)	68.39±0.46(8)	66.20±0.00(10)
musk	100.00±0.00(8.5) [+0.00%B, +13.26%A]	100.00±0.00(8.5)	100.00±0.00(8.5)	100.00±0.00(8.5)	97.00±2.27(20)	100.00±0.00(8.5)	92.21±6.32(22)	100.00±0.00(8.5)	100.00±0.00(8.5)	100.00±0.00(8.5)	100.00±0.00(8.5)
backdoor	89.90±0.72(1) [+0.81%B, +184.76%A]	89.01±0.00(3)	88.84±0.20(4)	45.70±12.50(10)	43.86±14.97(11)	46.54±1.41(9)	89.18±1.04(2)	14.20±0.63(16)	53.47±2.60(7)	4.83±0.10(29)	7.66±0.06(23)
speech	4.79±0.46(2) [-4.77%B, +48.69%A]	5.03±0.01(1)	4.19±0.41(3)	3.17±0.00(13)	2.94±0.31(19)	2.80±0.00(23)	3.38±0.50(8.5)	3.00±0.29(16)	3.15±0.00(14)	3.10±0.06(15)	2.78±0.00(25)
census	18.88±0.88(11) [-34.85%B, +13.58%A]	21.71±0.01(2)	14.57±0.52(19)	21.05±0.67(5)	16.01±2.97(13)	21.68±0.63(3)	21.20±0.61(4)	19.67±0.60(10)	13.71±0.42(23)	14.96±4.41(16)	20.32±0.66(7)
CIFAR10	16.84±2.25(19) [-29.56%B, -4.86%A]	21.01±0.00(5)	23.91±0.49(1)	19.91±0.00(8)	17.45±0.52(16)	19.62±0.00(10)	17.39±0.59(17)	19.55±0.08(11)	22.17±0.00(4)	19.98±0.07(7)	19.42±0.00(12)
SVHN	14.46±1.53(17) [-12.79%B, +3.58%A]	16.58±0.00(1)	16.20±0.65(2)	15.53±0.00(6)	14.67±0.33(16)	15.34±0.00(8)	15.60±0.31(5)	15.08±0.04(10)	15.97±0.00(4)	15.40±0.26(7)	15.00±0.00(12)
MVTec-AD	73.05±14.04(16) [-24.04%B, +0.15%A]	78.66±0.01(8)	96.17±0.55(1)	82.94±2.68(6)	73.25±1.79(15)	75.76±2.71(11)	89.46±2.23(3)	73.66±2.72(14)	75.79±2.95(9)	87.91±2.31(4)	73.03±2.82(17)
FashionMNIST	59.30±20.52(8) [-7.26%B, +26.39%A]	59.81±0.01(4)	42.48±2.48(21)	59.79±0.00(5)	52.19±2.78(16)	59.15±0.00(9)	63.08±0.96(3)	57.14±0.12(11)	63.61±0.00(2)	59.60±0.17(7)	56.53±0.00(13)
MNIST-C	56.98±22.01(1) [+9.26%B, +67.10%A]	51.24±0.00(5)	9.28±0.12(28)	47.42±0.00(6)	38.40±3.34(16)	46.20±0.00(8)	51.47±1.10(4)	41.78±0.12(11)	51.89±0.00(3)	46.89±0.14(7)	41.57±0.00(12)
yelp	15.94±1.27(6) [-12.13%B, +24.77%A]	18.14±0.00(1)	11.46±0.57(21)	16.35±0.00(2)	13.90±3.00(7)	16.03±0.00(5)	10.40±0.09(24)	12.80±0.02(16)	16.14±0.00(3)	9.96±0.05(27)	13.42±0.00(10)
imdb	9.40±0.25(11) [-51.27%B, -2.01%A]	9.23±0.00(13)	9.49±0.23(9)	8.97±0.00(18.5)	10.53±2.36(2)	8.92±0.00(21)	10.24±0.13(4)	8.70±0.02(28)	9.03±0.00(15.5)	9.79±0.03(6)	8.85±0.00(22)
agnews	26.03±5.84(1) [+0.50%B, +88.87%A]	24.88±0.00(4)	13.42±1.52(11)	17.35±0.00(5)	15.16±3.17(8)	16.68±0.00(6)	15.42±0.38(7)	11.85±0.03(18)	25.86±0.00(3)	12.30±0.06(15)	12.82±0.00(12)
20news	17.02±8.08(2) [-8.94%B, +32.93%A]	18.69±0.00(1)	13.97±1.61(8)	15.61±1.35(3)	13.51±0.84(11)	13.47±0.52(12)	14.62±0.88(7)	11.48±0.44(21)	15.04±0.67(5)	13.59±0.34(10)	11.83±0.52(17)
amazon	12.01±0.42(2) [-6.46%B, +11.75%A]	12.84±0.00(1)	10.82±0.20(15)	11.73±0.00(3)	11.10±1.46(8.5)	11.69±0.00(5)	10.19±0.08(22)	10.76±0.02(17)	11.04±0.00(13)	9.74±0.02(26)	11.06±0.00(11.5)
internetads	89.06±0.42(1) [+5.35%B, +80.44%A]	84.54±0.02(2)	81.72±3.79(3)	51.30±2.02(11)	38.29±5.22(24)	49.22±0.00(14)	60.03±1.39(7)	47.70±0.16(16)	50.43±0.00(12)	60.52±1.05(6)	48.15±0.00(15)

TABLE IV

ONE-SIDED WILCOXON SIGNED-RANK TEST p-VALUES FOR THE ALTERNATIVE THAT **uLEAD-TABPFN** OUTPERFORMS EACH METHOD, GROUPED BY FEATURE DIMENSIONALITY. COLUMNS SHOW SELECTED BASELINES USED IN THE SIGNIFICANCE SUMMARY. THE "WINS" COLUMN REPORTS, OUT OF ALL BASELINES (27 FOR ROC-AUC AND 28 FOR PR-AUC), HOW MANY uLEAD-TABPFN SIGNIFICANTLY OUTPERFORMS IN THAT GROUP. STATISTICAL SIGNIFICANCE IS INDICATED BY $^{*}p < 0.05$, $^{**}p < 0.01$, AND $^{***}p < 0.001$.

ROC-AUC	**Wins**	CAIT	DisentAD	FoMo-0D	DTE-NP	kNN	ICL	LOF	CBLOF	FeatBag	SLAD	DDPM
p-value ($d \leq 20$)	6/27	0.9553	0.0570	0.9677	0.9865	0.9876	0.7634	0.8202	0.8962	0.2031	0.3420	0.8739
p-value ($20 < d \leq 100$)	12/27	0.9955	0.0024**	0.5675	0.7525	0.7387	0.4493	0.1519	0.3830	0.1061	0.2754	0.0594
p-value ($d > 100$)	21/27	0.0789	0.0240*	0.0004***	0.2352	0.0886	0.1384	0.2163	0.0207*	0.2352	0.0320*	0.0038**
p-value (All)	18/27	0.9290	0.0003***	0.3966	0.9477	0.9233	0.4604	0.3328	0.3108	0.0464*	0.0357*	0.1137

PR-AUC	**Wins**	CAIT	DisentAD	DDAE	FoMo-0D	DTE-NP	kNN	ICL	LOF	SLAD	DDPM	OCSVM
p-value ($d \leq 20$)	10/28	0.9911	0.0197*	0.9997	0.9830	0.9903	0.9911	0.6580	0.6371	0.5279	0.8678	0.6051
p-value ($20 < d \leq 100$)	14/28	0.9506	0.0060**	0.9940	0.3830	0.7789	0.7659	0.4831	0.3047	0.3353	0.1733	0.2475
p-value ($d > 100$)	21/28	0.0603	0.0128*	0.8187	0.0017**	0.2163	0.0789	0.1147	0.1501	0.0368*	0.0093**	0.0093**
p-value (All)	17/28	0.9800	0.0001***	1.0000	0.6019	0.9630	0.9432	0.3890	0.3447	0.1805	0.3065	0.1783

TABLE V

FRIEDMAN/NEMENYI RANK-TEST SUMMARY OVER ALL 57 DATASETS. "BETTER" AND "WORSE" COUNT METHODS RANKED SIGNIFICANTLY ABOVE OR BELOW uLEAD-TABPFN UNDER THE NEMENYI POST-HOC TEST AT α=0.05.

Metric	Methods	Friedman p-value	uLEAD-TabPFN rank	Better	Worse
ROC-AUC	28	3.76×10^{-75}	10.84	0	11
PR-AUC	29	1.91×10^{-78}	11.18	0	11

APPENDIX F
DETAILED CASE STUDY: WILT DATASET

We present an end-to-end case study on the *Wilt* dataset to illustrate how uLEAD-TabPFN operates in practice, from latent representation learning to dependency-based scoring and interpretation. Wilt is a widely used benchmark for tabular anomaly detection derived from high-resolution remote sensing imagery, where diseased trees correspond to anomalous instances. The dataset consists of five continuous spectral and texture-related features extracted from multispectral and panchromatic image bands, including mean values of the green (Mean-G), red (Mean-R), and near-infrared (Mean-NIR) bands, as well as texture statistics from the panchromatic band (GLCM-Pan) and intensity variation measured by the standard deviation of the panchromatic band (SD-Pan). With a total of 4,819 samples (4,562 normal and 257 anomalous), its low dimensionality and well-defined physical feature semantics make Wilt particularly suitable for a detailed methodological analysis and interpretability study.

On the Wilt dataset, uLEAD-TabPFN achieves a ROC-AUC of 97.5% and a PR-AUC of 84.6%, substantially outperforming existing baseline methods, which typically achieve ROC-AUC values around 60% and PR-AUC values near 20%. Relative to the average baseline performance, this corresponds to improvements of 81% in ROC-AUC and 491% in PR-AUC.

The goal of this case study is to understand why uLEAD-TabPFN performs particularly well on this dataset. To this end, we compare uLEAD-TabPFN with the ablated variants introduced in the ablation study of the main paper. Specifically, uLEAD-Original, which operates directly on the original feature space without latent transformation, achieves 85% ROC-AUC and 47% PR-AUC. Replacing TabPFN with a CART-based dependency estimator (uLEAD-CART) obtains 92% ROC-AUC and 64% PR-AUC. uLEAD-TabPFN without residual-scale normalization attains similar ROC-AUC (97.4%) and PR-AUC (86.9%) to the full model.

These results indicate that the dominant performance gains on the Wilt dataset stem from the representation learning and dependency modeling enabled by TabPFN, while residual-scale normalization plays a less critical role in this particular setting. Consequently, the following analysis focuses on understanding how the latent space transformation contributes to improved anomaly detection performance.

a) Latent Space Analysis.: We first compare feature dependencies in the original and latent spaces using the Pearson correlation coefficient and the conditional coefficient of determination (R^2) estimated with TabPFN. Pearson correlation measures pairwise linear dependence, while conditional R^2 quantifies how well each feature can be predicted from the remaining features.

In the original feature space, the mean absolute Pearson correlation is 0.239, indicating weak pairwise dependencies. After projection into the latent space, this value increases to 0.531 (a 122.4% relative increase), with the standard deviation rising from 0.317 to 0.581, indicating amplified and more diverse dependency strengths. Consistent with this, the average conditional R^2 increases from 0.649 (std. 0.285) in the original space to 0.991 (std. 0.003) in the latent space, showing that the learned representation makes conditional relationships substantially more predictable and stable.

To examine how normal and anomalous samples are distributed in the original and latent spaces, we compare the two representations using multiple dimensionality reduction techniques, including PCA, t-SNE, and UMAP as shown in Figure 2.

Table VI reports complementary quantitative metrics characterizing geometric separation, density contrast, and neighborhood structure in both spaces. The silhouette score measures the degree of separation between normal and anomalous samples based on relative intra- and inter-class distances, with values near zero indicating substantial overlap. The distance to the nearest normal sample and the Local Outlier Factor

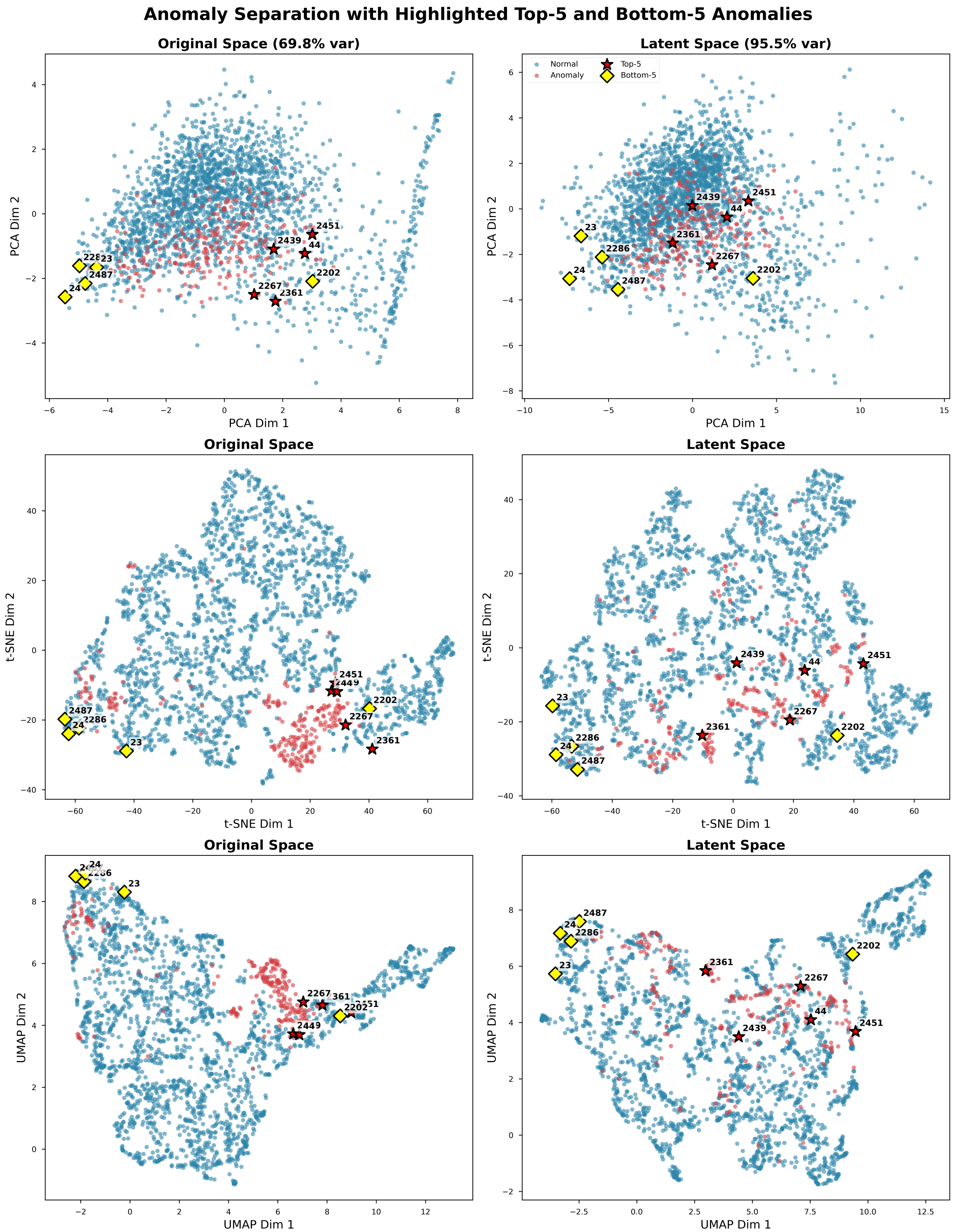

Fig. 2. Visualization of top-5 and bottom-5 anomalies by anomaly score in both the original and latent feature spaces of the Wilt dataset. Each panel shows PCA, t-SNE, and UMAP projections; normal samples are shown in gray, high-scoring anomalies in red, and low-scoring anomalies in blue.

(LOF) ratio capture marginal distance- and density-based distinguishability, while nearest-neighbor anomaly rates and k-NN purity quantify local neighborhood coherence among anomalous samples.

TABLE VI
COMPARISON OF GEOMETRIC AND NEIGHBORHOOD PROPERTIES IN THE ORIGINAL AND LATENT SPACES ON THE WILT DATASET.

Metric	Original	Latent	Interpretation
Average silhouette score (all samples)	-0.1402	0.0024	Normal and anomalous samples are highly overlapping in both spaces, with slightly improved separation in the latent space.
Average distance to nearest normal sample	0.3316	0.3387	The separation between normal and anomalous samples remains similar across spaces.
LOF ratio (anomaly / normal)	1.0383	1.0297	Density-based contrast remains limited and slightly weaker in the latent space.
Anomalies with anomaly as nearest neighbor	57.2%	71.2%	Anomalies exhibit stronger local neighborhood coherence in the latent space.
30-NN purity (fraction of anomalies)	0.2943	0.3926	Anomalies form more locally coherent neighborhoods in the latent space.

Figure 3 shows that normal and anomalous samples remain substantially overlapping in both the original and latent spaces, consistent with the near-zero silhouette scores reported in Table VI (from -0.140 in the original space to 0.002 in the latent space).

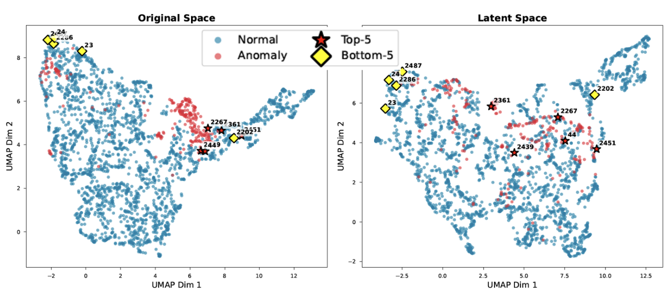

Fig. 3. UMAP visualization of the Wilt dataset in the original feature space (left) and the learned latent space (right). Normal samples are shown in gray and anomalous samples in color. Despite the latent transformation, normal and anomalous samples remain substantially overlapping; anomalous samples nevertheless form more locally coherent neighborhoods in the latent space.

Likewise, the average distance from anomalous samples to their nearest normal neighbors changes only marginally, from 0.332 to 0.339, and density-based contrast remains weak, with the LOF ratio slightly decreasing from 1.038 to 1.030. Together, these results indicate that the improved detection performance of uLEAD-TabPFN is not driven by global geometric separation or marginal distance- and density-based cues.

In contrast, both the visualization and neighborhood-level metrics reveal a clear structural change in the latent space. As reported in Table VI, the proportion of anomalies whose nearest neighbor is also anomalous increases from 57% to 71%, and the fraction of anomalies among the 30 nearest neighbors rises from 0.29 to 0.39, indicating stronger local neighborhood coherence among anomalous samples. In addition, the average silhouette score computed over anomalous samples remains positive in both spaces (0.331 in the original space and 0.275 in the latent space), whereas the corresponding scores for normal samples are negative (-0.167 and -0.013). Together, these results indicate that anomalous samples exhibit more coherent local neighborhood structure than normal samples in both representations, with stronger neighborhood-level coherence in the latent space despite slightly reduced global compactness.

These results indicate that in the original feature space, anomalies tend to form a single broad cluster that substantially overlaps with normal samples, which limits the effectiveness of distance-, density-, and marginal-distribution-based baseline methods. In contrast, Figure 3 illustrates that anomalous samples in the latent space are organized into multiple locally coherent neighborhoods rather than a single globally compact cluster. Although anomalies are not more geometrically separated from normal samples, the latent representation restructures anomalous instances according to shared dependency violations. This organization aligns naturally with the dependency-based scoring of uLEAD-TabPFN, where anomaly detection is driven by conditional inconsistency rather than marginal distance or density.

b) Anomaly Interpretation.: uLEAD-TabPFN exposes its dimension-wise conditional Gaussian NLL contributions, each combining a standardized dependency deviation with a conditional log-scale term. For diagnostic interpretation, we separately inspect the standardized-deviation component to identify which conditional relationships are most strongly violated. This residual-based diagnostic does not constitute an exact decomposition of the complete NLL because it excludes the log-scale term.

Because the latent representation is linear, the absolute encoder weights provide a transparent heuristic for distributing this latent standardized-deviation evidence over the original input features. The resulting residual-based attribution indicates which inputs are associated with the violated latent relationships, but it is neither a causal attribution nor an exact decomposition of the full NLL, whose conditional mean and scale are nonlinear functions of the latent variables.

We analyze a representative top-ranked anomaly from the Wilt dataset to illustrate this diagnostic. Its squared standardized deviations are substantially larger for Z_0, Z_1, Z_2, and Z_4 than for Z_3 (Table VIII). Mapping this evidence through the absolute linear-encoder weights yields the residual-based feature associations reported in Table VII; the strongest correspond to near-infrared reflectance (Mean-NIR) and panchromatic intensity variation (SD-Pan).

We provide a numerical breakdown for the highest-ranked anomaly (Index: 2361). Table VII reports its raw and normalized feature values together with the residual-based attribution. Table VIII reports the per-dimension conditional quantities. The NLL row includes the conventional constant $\frac{1}{2}\log(2\pi)$, which is omitted from the implemented score because it does not affect rankings; the squared standardized-deviation row isolates the residual component of the NLL.

Overall, this case study links the composite conditional NLL to a residual-based diagnostic of latent dependency violations and their heuristic associations with original features.

TABLE VII
RAW VALUES, NORMALIZED VALUES, AND RESIDUAL-BASED ATTRIBUTION SCORES FOR ANOMALY #1.

Metric	GLCM_Pan	Mean_G	Mean_R	Mean_NIR	SD_Pan
Raw Value	150.181	225.000	145.682	540.500	13.908
Raw Percentile	92.4%	56.8%	86.7%	52.8%	14.7%
Normalized Value	2.499	0.269	3.714	0.129	-1.509
Residual Attribution	5.181	3.649	3.978	6.193	5.240

TABLE VIII
PER-LATENT-DIMENSION ANALYSIS FOR ANOMALY #1, INCLUDING CONVENTIONAL GAUSSIAN NLL VALUES AND THEIR SQUARED STANDARDIZED-DEVIATION COMPONENTS.

Metric	Z_0	Z_1	Z_2	Z_3	Z_4
Latent Value	-1.352	-0.627	-1.373	-0.024	0.247
Latent Percentile	12.9%	26.6%	19.4%	61.4%	54.6%
Conditional Mean	-0.284	-2.146	0.569	0.316	2.006
Residual	1.068	1.519	1.942	0.340	1.760
Conditional Variance	1.000	1.000	1.000	1.000	1.000
NLL (incl. constant)	1.489	2.072	2.805	0.977	2.467
Squared Std. Deviation	1.140	2.306	3.771	0.116	3.096

APPENDIX G
LATENT-DIMENSION SENSITIVITY

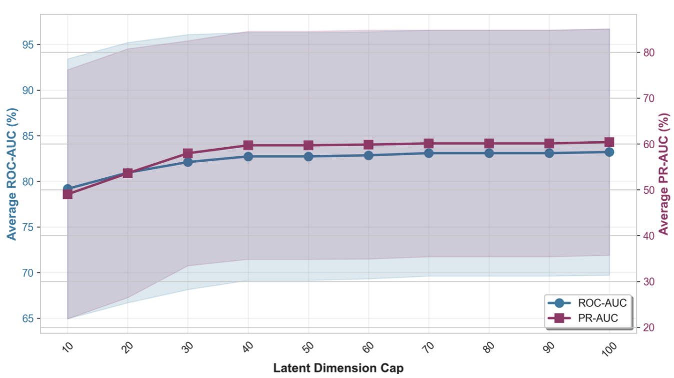

Fig. 4. Sensitivity of uLEAD-TabPFN to the latent dimension cap p on datasets with $d \leq 100$. The blue curve (left axis) shows the average ROC-AUC and the red curve (right axis) the average PR-AUC as functions of p.

uLEAD-TabPFN is deliberately designed with a minimal set of hyperparameters, most of which are fixed across all experiments (see the main paper for details) to avoid dataset-specific tuning. We therefore focus the sensitivity analysis on the primary structural hyperparameter, the latent dimensionality p, restricting it to datasets with $d \leq 100$ to avoid confounding the effect of p with the aggressive dimensionality reduction that would be required for datasets with $d > 100$. Figure 4 reports the average ROC-AUC and PR-AUC over these datasets as p varies: performance improves rapidly as p increases from small values and stabilizes around $p \approx 40$, with only marginal gains for larger values up to $p = 100$. These results indicate that uLEAD-TabPFN does not require fine-grained tuning of the latent dimensionality and maintains stable performance across a broad range of values.

APPENDIX H
RCS-BUDGET SENSITIVITY

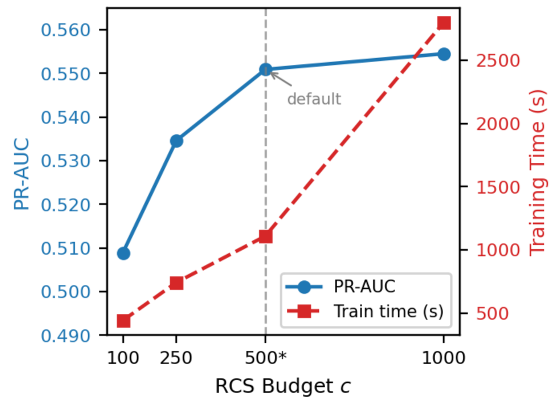

Fig. 5. Sensitivity to the RCS budget under the final evaluation protocol. Increasing the context budget improves PR-AUC but substantially increases training time; the default budget $c = 500$ provides a practical trade-off.

Figure 5 reports sensitivity to the RCS budget. Increasing the budget from $c = 100$ to $c = 1000$ improves PR-AUC from 0.5089 to 0.5545, while ROC-AUC remains stable across the tested budgets, with values of 0.8093, 0.8080, 0.8078, and 0.8071 for $c = 100$, 250, 500, and 1000, respectively. However, training time increases sharply from 442.1 s to 2797.4 s overall, with the steepest rise between $c = 500$ and $c = 1000$: doubling the budget more than doubles the training time (1109.8 s to 2797.4 s), consistent with the increased cost of in-context TabPFN evaluation over larger context sets. The default budget $c = 500$ achieves nearly the same PR-AUC as $c = 1000$ (0.5509 vs. 0.5545) at substantially lower training cost, providing a practical accuracy-efficiency trade-off.

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Algorithm 1 Training uLEAD-TabPFN

Require: Normal training set $\mathcal{X}_{\text{train}} = \{\mathbf{x}_i\}_{i=1}^N$, context budget c, latent dimension cap p
Require: Frozen TabPFN conditional predictor $\text{TabPFN}(\cdot)$
Ensure: Trained encoder f_θ, trained variance networks $\{v_{\phi_j}\}_{j=1}^p$, context set $\mathcal{C}$, normalization stats $(\mu_\mathcal{C}, \sigma_\mathcal{C})$

Construct Representative Context Set:
$\mathcal{C} \leftarrow \text{BuildContext}(\mathcal{X}_{\text{train}}, c)$ ▷ e.g., KMeans-based selection
Compute $(\mu_\mathcal{C}, \sigma_\mathcal{C})$ on $\mathcal{C}$
Normalize $\tilde{\mathbf{x}} \leftarrow (\mathbf{x} - \mu_\mathcal{C}) \oslash \sigma_\mathcal{C}$ for all $\mathbf{x} \in \mathcal{X}_{\text{train}} \cup \mathcal{C}$
Train encoder f_θ:
Initialize encoder parameters θ ▷ e.g., $f_\theta(\tilde{\mathbf{x}}) = W\tilde{\mathbf{x}} + b$
for each epoch **do**
  for each mini-batch $\mathcal{B} \subset \mathcal{X}_{\text{train}}$ **do**
    Compute latent batch $Z_\mathcal{B} \leftarrow \{\mathbf{z}_i = f_\theta(\tilde{\mathbf{x}}_i) \mid \mathbf{x}_i \in \mathcal{B}\}$
    Compute context latents $Z_\mathcal{C} \leftarrow \{\mathbf{z} = f_\theta(\tilde{\mathbf{x}}) \mid \mathbf{x} \in \mathcal{C}\}$
    $L_{\text{dep}} \leftarrow 0$
    for $j = 1$ to p **do**
      $\hat{\mu}_{ij} \leftarrow \text{TabPFNMean}(j; \mathbf{z}_{i,-j}, Z_\mathcal{C})$ for each $\mathbf{z}_i \in Z_\mathcal{B}$
      $L_{\text{dep}} \leftarrow L_{\text{dep}} + \frac{1}{|\mathcal{B}|}\sum_{\mathbf{z}_i \in Z_\mathcal{B}} (z_{ij} - \hat{\mu}_{ij})^2$
    end for
    $L_{\text{dep}} \leftarrow \frac{1}{p} L_{\text{dep}}$
    $L_{\text{rec}} \leftarrow \frac{1}{|\mathcal{B}|}\sum_{\mathbf{x}_i \in \mathcal{B}} \|\tilde{\mathbf{x}}_i - W^\top \mathbf{z}_i\|_2^2$ ▷ optional
    $L_{\text{enc}} \leftarrow \lambda_{\text{dep}} L_{\text{dep}} + \lambda_{\text{rec}} L_{\text{rec}}$
    Update θ using L_{enc} (TabPFN frozen)
  end for
end for
Train variance networks $\{v_{\phi_j}\}$ on $\mathcal{X}_{\text{train}}$:
for $j = 1$ to p **do**
  Initialize ϕ_j
  for each epoch **do**
    for each mini-batch $\mathcal{B} \subset \mathcal{X}_{\text{train}}$ **do**
      Compute $Z_\mathcal{B} \leftarrow \{\mathbf{z}_i = f_\theta(\tilde{\mathbf{x}}_i) \mid \mathbf{x}_i \in \mathcal{B}\}$ and $Z_\mathcal{C} \leftarrow \{\mathbf{z} = f_\theta(\tilde{\mathbf{x}}) \mid \mathbf{x} \in \mathcal{C}\}$
      $\hat{\mu}_{ij} \leftarrow \text{TabPFNMean}(j; \mathbf{z}_{i,-j}, Z_\mathcal{C})$ for $\mathbf{x}_i \in \mathcal{B}$
      $\log \hat{\sigma}_{ij}^2 \leftarrow v_{\phi_j}(\mathbf{z}_{i,-j})$ for $\mathbf{x}_i \in \mathcal{B}$
      $L_{\text{var}} \leftarrow \frac{1}{2|\mathcal{B}|}\sum_{\mathbf{x}_i \in \mathcal{B}} \left[\frac{(z_{ij} - \hat{\mu}_{ij})^2}{\hat{\sigma}_{ij}^2 + \epsilon} + \log \hat{\sigma}_{ij}^2\right]$
      Update ϕ_j using L_{var} (encoder and TabPFN frozen)
    end for
  end for
end for
return f_θ, $\{v_{\phi_j}\}_{j=1}^p$, $\mathcal{C}$, $(\mu_\mathcal{C}, \sigma_\mathcal{C})$

Algorithm 2 Inference with uLEAD-TabPFN

Require: Test sample $\mathbf{x}$, context set $\mathcal{C}$, normalization stats $(\mu_{\mathcal{C}}, \sigma_{\mathcal{C}})$

Require: Trained encoder f_θ, trained variance nets $\{v_{\phi_j}\}_{j=1}^{p}$, frozen TabPFN

Ensure: Anomaly score $s(\mathbf{x})$

Normalize $\tilde{\mathbf{x}} \leftarrow (\mathbf{x} - \mu_{\mathcal{C}}) \oslash \sigma_{\mathcal{C}}$

Compute $\mathbf{z} \leftarrow f_\theta(\tilde{\mathbf{x}})$

Compute $Z_{\mathcal{C}} \leftarrow \{f_\theta(\tilde{\mathbf{x}}) \mid \tilde{\mathbf{x}} \in \tilde{\mathcal{C}}\}$

$s(\mathbf{x}) \leftarrow 0$

for $j = 1$ to p **do**

  $\hat{\mu}_j \leftarrow \mathrm{TabPFNMean}(j; \mathbf{z}_{-j}, Z_{\mathcal{C}})$

  $\log \hat{\sigma}_j^2 \leftarrow v_{\phi_j}(\mathbf{z}_{-j})$

  $s(\mathbf{x}) \leftarrow s(\mathbf{x}) + \frac{1}{2}\left[\frac{(z_j - \hat{\mu}_j)^2}{\hat{\sigma}_j^2 + \epsilon} + \log \hat{\sigma}_j^2\right]$

end for

$s(\mathbf{x}) \leftarrow \frac{1}{p} s(\mathbf{x})$

return $s(\mathbf{x})$